Abstract

Large Language Models (LLMs) have enabled powerful general-purpose dialogue systems, but their behaviour degrades in long, multi-topic, or entangled conversations. This literature review synthesizes research on structured reasoning (Chain-of-Thought and its extensions), planning-based inference (Tree-of-Thought and Graph-of-Thought), long-context limitations, conversation disentanglement, hierarchical dialogue modelling, memory-augmented architectures, retrieval-augmented generation, multi-agent communication frameworks, and recent work specifically targeting multi-turn interaction reliability and training. The aim is to establish a conceptual and empirical basis for *Subchat Trees*, a proposed hierarchical, multi-branch conversational architecture that explicitly structures interaction into topic-specific sub-conversations. The review identifies a gap: existing work structures *reasoning*, *memory*, or *agent interaction*, and some recent work introduces branching for *multi-turn training*, but there remains limited emphasis on a user-facing, proactive mechanism to prevent topic entanglement and long-context degradation in single-user LLM chats. Subchat Trees are positioned as a system-level response to this gap.

1 Introduction

Large Language Models (LLMs) such as GPT-style models and their open-source counterparts have rapidly become the backbone of interactive AI systems. They are deployed in chat interfaces for programming assistance, education, content creation, and open-domain dialogue. Despite these successes, practitioners and users frequently observe failures in extended, multi-topic conversations: the model forgets earlier details, confuses distinct topics, mixes separate tasks, or relies on outdated context. These pathologies suggest that, even as context windows grow, unstructured linear transcripts remain an ill-suited substrate for robust reasoning and dialogue management.

In parallel, several lines of research indicate that *structure*—in reasoning, memory, and interaction—can significantly improve LLM capabilities. Chain-of-Thought (CoT) prompting demonstrates that explicitly representing intermediate reasoning steps boosts performance on arithmetic, commonsense, and symbolic tasks (Wei et al., 2022). Self-Consistency further improves CoT by aggregating multiple reasoning paths (Wang et al., 2023). Tree-of-Thought (ToT) and Graph-of-Thought (GoT) extend this idea to tree- and graph-based search over thoughts (Yao et al., 2023; Besta et al., 2023). At the same time, empirical work shows that LLMs fail to reliably use long contexts, especially for information in the middle of an input sequence (Liu et al., 2023). In the dialogue domain, conversation disentanglement research documents how multi-party chat streams contain interleaved conversations that must be separated for downstream use (Kummerfeld and et al., 2019). Hierarchical dialogue models (Serban et al., 2016), memory systems such as MemGPT (Packer et al., 2023), retrieval-augmented generation (RAG) (Lewis et al., 2020), and multi-agent communication frameworks like CAMEL (Li et al., 2023) each add structure at different levels of the LLM stack.

More recently, multi-turn interaction has become a first-class research focus. Li et al. synthesize evaluation settings, benchmarks, and enhancement approaches for multi-turn LLM interaction, emphasizing challenges such as maintaining context, coherence, and robustness over long dialogues (Li et al., 2025). Complementing this survey perspective, Savage proposes *Savage Conversation Forests* (SCF), a branching training architecture for fine-tuning LLMs on multi-turn tasks (demonstrated in medical interviews), directly arguing that linear fine-tuning schemes fail to capture inter-turn dependencies (Savage, 2025). Together, these works reinforce the central motivation of this review: multi-turn conversation is not merely "single-turn repeated," and explicit structural interventions may be necessary at both training-time and interface-time.

This review asks: **What does existing literature tell us about the importance and form of structure in LLM-based dialogue systems, and where are the gaps that motivate a hierarchical, multi-branch conversational architecture such as Subchat Trees?**

Following recommendations for literature reviews (e.g. **?**), this paper does not simply list sources, but instead synthesizes and critically evaluates them with respect to a unifying question. Section 2 surveys foundational work on end-to-end dialogue modelling. Section 3 examines structured reasoning methods (CoT, Self-Consistency, ToT, GoT). Section 4 addresses long-context limitations, memory architectures, and retrieval-based augmentation. Section 5 reviews conversation disentanglement. Section 6 reviews recent work directly targeting multi-turn interaction evaluation and training (including surveys and branching fine-tuning). Section 7 covers multi-agent and role-based frameworks. Section 8 synthesizes findings across strands, and Section 9 identifies the research gap and positions Subchat Trees.

2 Background: End-to-End Dialogue Modelling

Before the rise of LLMs, neural dialogue systems were predominantly trained as end-to-end generative models on conversational corpora. Serban et al. introduced a hierarchical recurrent encoder–decoder (HRED) architecture that models dialogues at two levels: a sequence of tokens within utterances, and a sequence of utterances within a conversation (Serban et al., 2016). Their generative dialogue model decomposes the probability of a conversation into a product over utterances, each conditioned on a latent dialogue state that is updated in a higher-level recurrent network. This design allows the model to maintain a representation of dialogue context over multiple turns while generating each utterance token-by-token.

The HRED work is important for two reasons in the present context. First, it empirically supports the intuition that *hierarchical* structure improves modelling of dialogue coherence relative to flat sequence models. Second, it demonstrates that structure can be applied at different levels: in HRED, the structure is internal to the model (latent dialogue states and utterance-level encoding), not external in the user interface or conversation management layer.

However, several limitations reduce the direct applicability of HRED to modern LLM-based systems. Training is corpus-specific and domain-dependent, and the model does not provide mechanisms for users to explicitly manage topics, branch conversations, or prevent entanglement. Subchat Trees can be seen as an architectural analogue that exposes a form of hierarchical structure to users, rather than only within model parameters.

3 Structured Reasoning with Large Language Models

3.1 Chain-of-Thought Prompting

Chain-of-Thought prompting (Wei et al., 2022) is a landmark technique demonstrating that prompting large LLMs with few-shot exemplars containing intermediate reasoning steps can dramatically improve performance. Instead of standard input–output pairs, CoT uses triples of the form ⟨input, chain-of-thought, answer⟩, where the chain-of-thought is a natural-language explanation or derivation leading to the solution. Experiments on math word problems, commonsense reasoning, and symbolic tasks show large gains, especially for models with $\gtrsim$100B parameters.

CoT has several properties relevant to Subchat Trees. It shows that:

- the *form* and *structure* of the prompt significantly influence reasoning performance;
- multi-step reasoning can be elicited through appropriately structured examples;
- structured intermediate representations can also provide interpretability and opportunities for debugging.

However, CoT operates within a single prompt or question. It does not address multi-turn dialogue, topic switching, or context management over time.

3.2 Self-Consistency over Chains of Thought

Wang et al. extend CoT with a decoding strategy called Self-Consistency (Wang et al., 2023). Instead of relying on a single greedy reasoning path, Self-Consistency:

1. samples multiple diverse chains of thought using temperature-based decoding;
2. extracts the final answer from each chain;

3. aggregates answers via majority vote to select the most consistent output.

This highlights that exploring *multiple reasoning branches* and aggregating over them can improve robustness. The connection to Subchat Trees is conceptual: branching can be applied not only within a single answer, but across extended interactions.

3.3 Tree-of-Thought: Search over Thoughts

Tree-of-Thought (ToT) generalizes CoT into an explicit tree search framework for problem solving with LLMs (Yao et al., 2023). Instead of a single chain, ToT treats partial solutions ("thoughts") as nodes in a tree, exploring candidate steps via breadth-first or depth-first search with evaluation heuristics. ToT reinforces two ideas relevant to Subchat Trees: (i) maintaining multiple partial solutions can be beneficial, and (ii) trees are a natural representation for exploration and backtracking.

3.4 Graph-of-Thought: Generalizing to Graphs

Graph-of-Thought (GoT) models reasoning as an arbitrary directed graph rather than a tree (Besta et al., 2023). Thoughts are vertices and edges encode dependencies, allowing aggregation, refinement loops, and more complex topologies. While Subchat Trees are tree-structured at the interface level, GoT motivates the idea that branches can still be related via cross-links or shared subresults.

3.5 Synthesis and Critique

CoT, Self-Consistency, ToT, and GoT converge on a central conclusion: *structured reasoning primitives dramatically increase LLM capability*. However, these methods largely operate at the scale of *single problems* rather than extended, multi-topic conversations. They do not prescribe how to manage or segment dialogue history across tasks or time, and their structures are typically ephemeral rather than persistent conversational state. Subchat Trees aim to extend the benefits of branching and structure from reasoning-time to interaction-time.

4 Context, Memory, and Retrieval

4.1 Limitations of Long Contexts: “Lost in the Middle”

Liu et al. analyse how LLMs use long contexts in retrieval and integration tasks (Liu et al., 2023). They find a U-shaped performance curve: models are most accurate when relevant information appears at the beginning or end, and least accurate when it appears in the middle. This implies that simply increasing context windows is insufficient; long, unstructured transcripts naturally bury important information. Subchat Trees address this by keeping per-branch histories compact and topic-specific.

4.2 Memory Hierarchies: MemGPT

MemGPT treats the LLM context window as a limited memory resource, using an explicit memory hierarchy inspired by virtual memory (Packer et al., 2023). It distinguishes main context from external storage and uses operations to page information in and out. This mitigates context-length limits but does not by itself solve topic entanglement; Subchat Trees are complementary, providing a natural unit (a branch) for scoping memory.

4.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) integrates document retrieval with generation to improve knowledge-intensive tasks (Lewis et al., 2020). RAG primarily addresses factual grounding and knowledge access rather than interaction structure. However, a Subchat Tree node could be RAG-augmented, and retrieval could be scoped per branch.

4.4 Discussion

These works motivate the idea that unstructured context is problematic and that managed memory and retrieval help. Yet they leave open the question of *how to structure the conversation itself*. Subchat Trees propose a structural layer above memory and retrieval: the layout and segmentation of the dialogue.

5 Conversation Disentanglement

Kummerfeld et al. provide a large-scale dataset and benchmarks for conversation disentanglement in IRC logs (Kummerfeld and et al., 2019). Interleaved multi-party messages require recovering reply structure graphs, and the authors show that disentanglement is difficult and error-prone. This is directly relevant to Subchat Trees: disentanglement is reactive (recovering structure after entanglement), whereas Subchat Trees advocate proactive structuring at creation time.

6 Multi-Turn Interaction: Evaluation and Branching Training

6.1 Survey Perspective: Taxonomies and Challenges in Multi-Turn LLM Interaction

Li et al. survey multi-turn interactions with LLMs, organizing the space across evaluation scenarios (e.g., instruction-following over multiple turns, roleplay, healthcare, education, safety/adversarial settings) and across enhancement methods (e.g., contextual learning, fine-tuning, reinforcement learning, memory/retrieval integration, and agent-based approaches) (Li et al., 2025). A key implication for Subchat Trees is that multi-turn success depends not only on raw context length, but on robustness to conversational phenomena such as drift, coherence breaks, and accumulating error over turns. The survey's broad view supports treating multi-turn interaction as a distinct capability profile, suggesting that conversation design itself should be considered an intervention point, not only model training or prompting.

6.2 Branching for Multi-Turn Fine-Tuning: Savage Conversation Forests

Savage proposes *Savage Conversation Forests* (SCF), a fine-tuning framework that explicitly introduces branching conversation continuations during training for multi-turn dialogue (Savage, 2025). The motivating claim is that common fine-tuning methods (e.g., SFT-style single-turn completion training or preference optimization on isolated prompts) do not efficiently teach how early conversational actions shape later outcomes. SCF generates multiple possible continuations at each turn, enabling richer interdependent training signals and improved downstream performance on multi-turn tasks (demonstrated in diagnostic medical interviewing).

SCF is especially relevant to Subchat Trees because it provides an example where branching is not only a reasoning-time trick (as in ToT) but a core mechanism for learning multi-turn behavior. Conceptually, SCF strengthens the argument that the "right" representation of conversation may be tree-like rather than strictly linear. Subchat Trees extend this branching idea from the training process to the user-facing interaction layer: instead of branching only in the training data or internal rollouts, the conversational interface itself can expose and manage branches.

6.3 Implications for Subchat Trees

Taken together, the survey view (Li et al., 2025) and branching fine-tuning view (Savage, 2025) suggest that:

- multi-turn interaction should be evaluated and optimized as its own category, not assumed to follow from single-turn improvements;
- branching is a plausible structural primitive for handling multi-turn dependencies;
- there is an opportunity for system-level approaches that prevent entanglement by design, rather than relying solely on stronger models or post hoc recovery.

7 Multi-Agent and Role-Based Interaction

CAMEL introduces role-playing among multiple LLM agents and protocols for cooperative problem solving (Li et al., 2023). Explicit interaction structure (roles, turn-taking, critic agents) can improve coherence and task completion. From the Subchat Trees perspective, CAMEL shows that structuring interaction changes outcomes; Subchat Trees apply this idea to a single-user setting by offering role- or task-specific branches rather than multiple distinct agents.

8 Synthesis Across Strands

Across structured reasoning, long-context analysis, memory and retrieval, disentanglement, multi-turn interaction research, and multi-agent interaction, several themes emerge.

First, **structure matters**. CoT, Self-Consistency, ToT, and GoT show that embedding structure into reasoning improves performance. HRED and MemGPT extend structure to dialogue modelling and memory. RAG structures knowledge access. Multi-agent frameworks structure interaction protocols. Recent multi-turn work further emphasizes that multi-turn dialogue is a special regime with distinct evaluation and training needs (Li et al., 2025), and that branching can improve multi-turn learning signals (Savage, 2025).

Second, **unstructured linear context is fragile**. Lost-in-the-middle demonstrates uneven accessibility in long sequences; disentanglement shows that interleaving threads creates confusion and requires imperfect recovery. Multi-turn research highlights compounding failures across turns, motivating proactive structuring rather than purely scaling context length.

Third, **existing work is mostly component-level**. Most methods intervene at prompting, decoding, memory/retrieval, evaluation, or training-time branching. They rarely propose a persistent, user-facing conversational structure as the primary interface abstraction.

These observations suggest an opening for interaction architectures that expose branching and segmentation to users, keeping contexts compact, preventing entanglement, and supporting parallel exploration.

9 Research Gap and Positioning of Subchat Trees

The reviewed literature supports several claims that motivate Subchat Trees:

(a) LLMs benefit from structured reasoning and managed memory.

(b) Long, unsegmented contexts are difficult for models to use reliably.

(c) Real-world conversations often involve multiple overlapping threads, which are hard to disentangle after the fact.

(d) Multi-turn interaction is a distinct capability area with dedicated evaluation and enhancement strategies (Li et al., 2025).

(e) Branching architectures can improve multi-turn training signals and outcomes (Savage, 2025).

However, none of the surveyed works propose a *user-facing, hierarchical conversation architecture* that:

- allows users to branch off sub-conversations from any message;
- isolates context per branch to prevent cross-topic contamination;
- supports parallel exploration of alternative interpretations or solutions;
- integrates with memory and retrieval systems to keep each branch compact and relevant.

This constitutes the core research gap. Current systems still default to a flat, chronological transcript as the organizing principle for conversations, despite ample evidence that this representation is suboptimal for both models and users.

Subchat Trees are positioned as a response to this gap. Conceptually:

> A hierarchical, tree-structured interface for LLM-based dialogue, where each node represents a message and each branch represents a sub-conversation

with its own local context. Branches can be created, navigated, and pruned explicitly, allowing multi-topic and multi-threaded interactions without entanglement.

In light of the literature:

- Subchat Trees operationalise ToT-like branching, but at the *dialogue* level rather than the single-problem level.
- They complement MemGPT and RAG by providing a natural unit (a branch) for memory and retrieval scoping.
- They address lost-in-the-middle by keeping per-branch histories short and focused.
- They proactively prevent entanglement, aligning with insights from conversation disentanglement.
- They align with recent multi-turn emphasis (surveyed by Li et al.) and echo branching motivations seen in SCF, but reposition branching as an interface and conversation-management primitive rather than only a training-time technique.

Future empirical work should evaluate whether Subchat Trees improve task performance, user satisfaction, and error rates in multi-topic and long-running interactions compared to linear chats.

10 Conclusion

This literature review has examined influential work on structured reasoning, memory and retrieval, long-context behaviour, conversation disentanglement, hierarchical dialogue modelling, multi-agent interaction, and recent work specifically targeting multi-turn interaction reliability and training. Collectively, they support the thesis that structure—in both internal reasoning and external interfaces—is essential for robust, scalable conversational AI.

Yet current systems largely rely on flat conversational transcripts and provide limited affordances for users to structure their own interactions. Subchat Trees extend the principle of structured reasoning and branching to the level of conversation design, proposing a tree-based architecture for multi-topic, multi-stage dialogue. The reviewed literature motivates this direction and provides conceptual tools and empirical baselines against which Subchat Trees can be evaluated.

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